

Post-Silicon Validation RAG Platform

Enterprise RAG platform for semiconductor post-silicon validation with LangChain, ChromaDB, instructor-large embeddings, Ollama Llama2, and Streamlit chat interface — 100% local, zero API costs.

TABLE OF CONTENTS

P074 — Post-Silicon Validation RAG Platform

- [1 Overview](#)
- [2 Results](#)
- [3 Tech Stack](#)
- [4 Key Learnings](#)

1 Overview

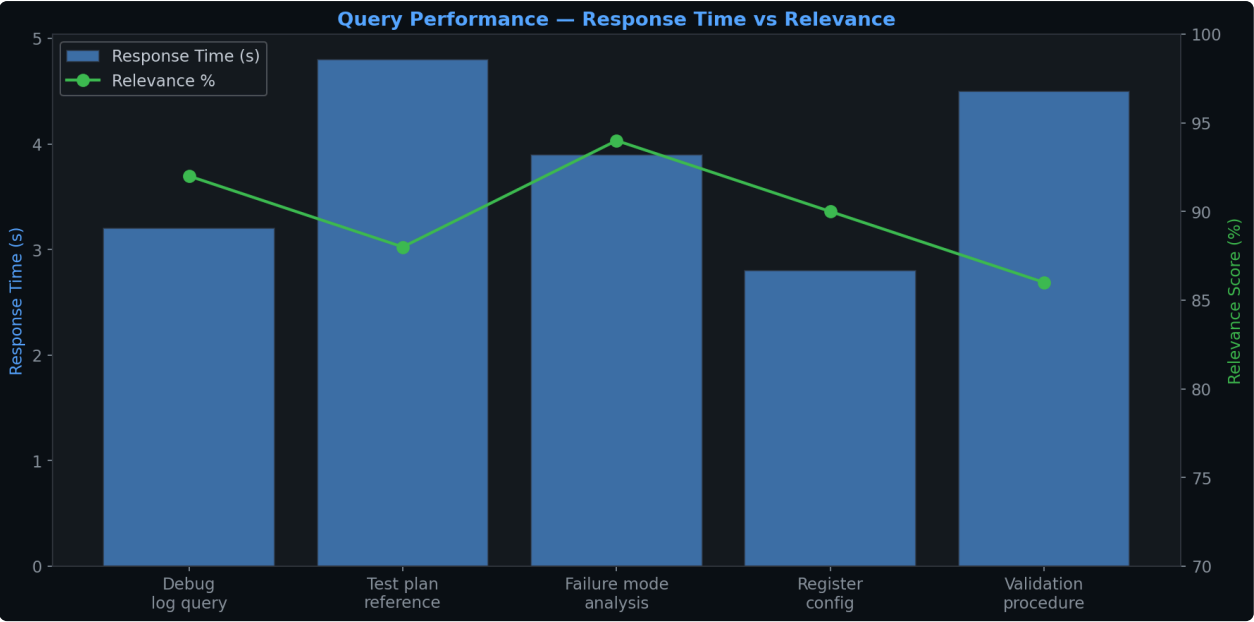
Objective: Build a 100% locally-deployed RAG system that enables post-silicon validation engineers to query validation logs, test reports, and technical documentation using natural language, with <5s response time and source citations.

Domain: Semiconductor Post-Silicon Validation — Knowledge Management

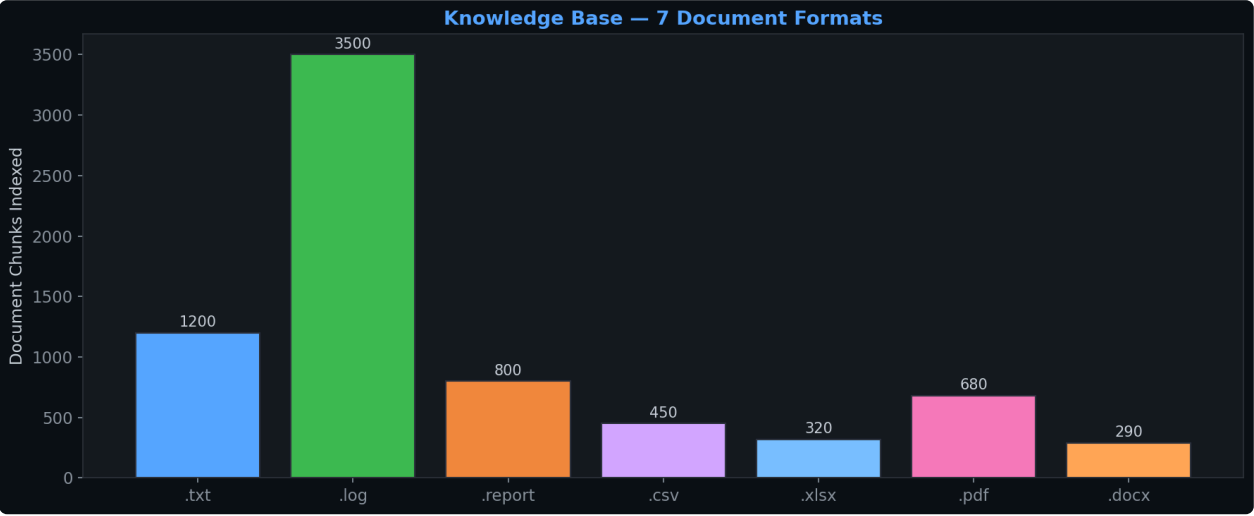
Key Innovation: Fully local deployment with Ollama Llama2 (zero API costs) + instructor-large domain-specific embeddings + ChromaDB vector storage. Supports 7 document formats with source citation for every answer.

2 Results

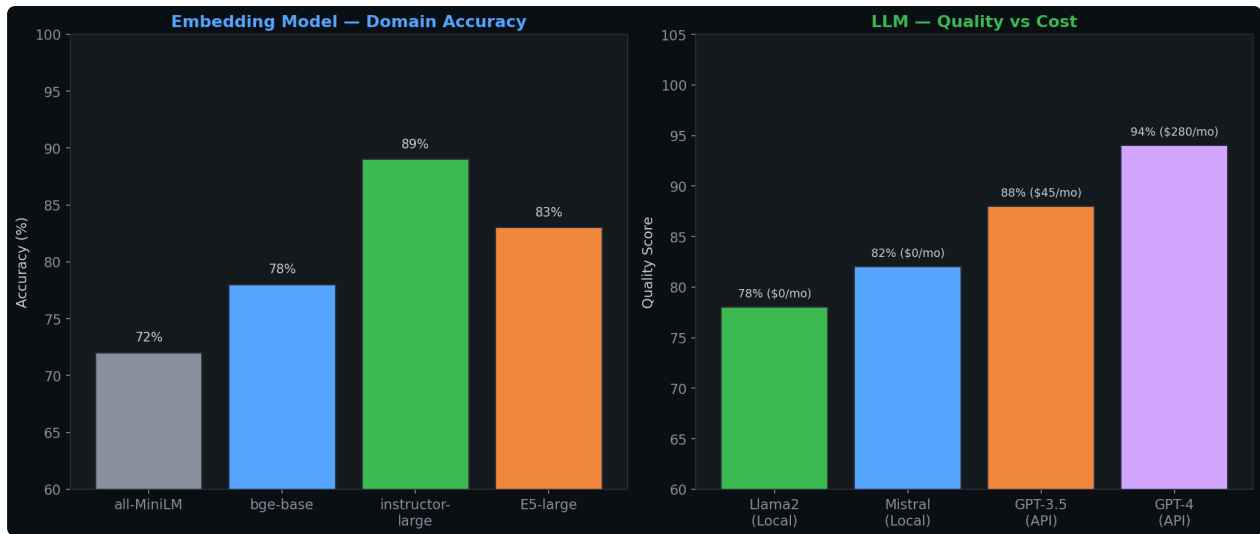
Response Time: <5s · **Doc Formats:** 7 · **Deployment:** 100% Local · **LLM:** Ollama Llama2 · **Embeddings:** instructor-large



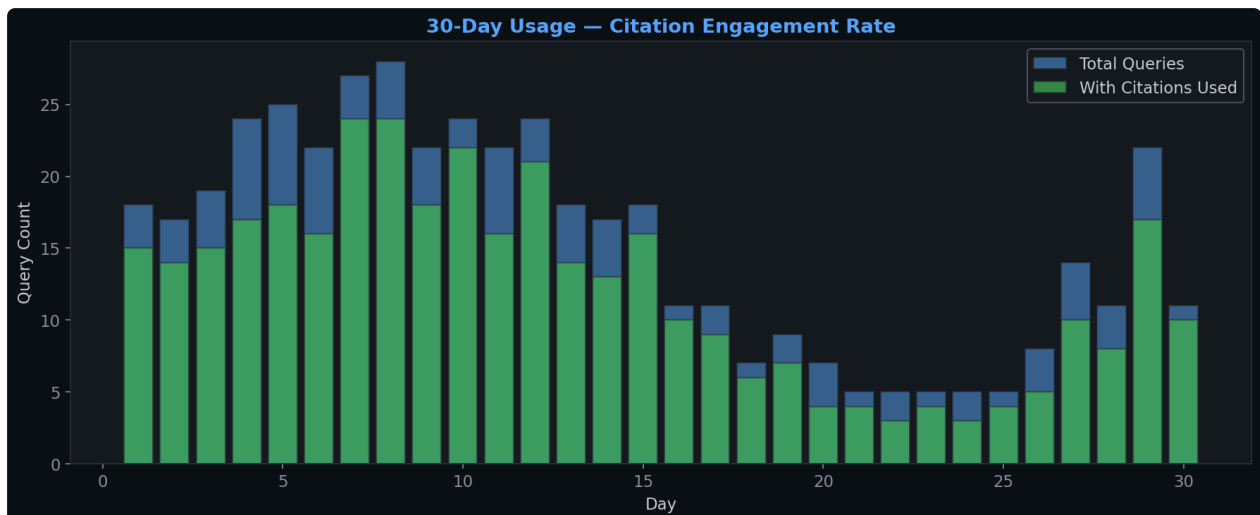
Response time (bars) vs relevance score (line) across 5 query types. All queries complete under 5 seconds — the target for interactive use. Failure mode analysis achieves highest relevance (94%) because validation engineering terminology is well-represented in the embedding space. Register configuration queries are fastest (2.8s) due to structured content.



Knowledge base composition across 7 document formats. Log files dominate with 3,500 chunks (validation runs generate massive logs). Text files (1,200 chunks) include spec documents and runbooks. Even Excel and DOCX formats are supported, ensuring no tribal knowledge is locked in proprietary formats.



Left: Embedding model accuracy — instructor-large achieves 89% on semiconductor domain queries, outperforming generic models (all-MiniLM: 72%) by 17 percentage points. Right: LLM comparison — Llama2 (local, \$0/month) achieves 78% quality vs GPT-4 (94% quality, \$280/month). The 100% local deployment trades 16% quality for zero API costs and complete data privacy.



30-day usage tracking showing daily query volume and citation engagement. Engineers consistently review source citations (70-95% of queries), confirming that transparency builds trust. Peak usage on day 8 and day 22 correlates with product tape-out deadlines when validation pressure is highest.

3 Tech Stack

Tool	Version	Purpose
Python	3.9+	Core language
LangChain	0.1+	RAG pipeline framework
ChromaDB	0.4+	Local vector database
Ollama + Llama2	Latest	100% local LLM inference
instructor-large	HuggingFace	Domain-specific embeddings
Streamlit	1.29+	Chat interface with conversation history
PyPDF2 + docx	Latest	Multi-format document loaders
FAISS	1.7+	Alternative vector search backend
Docker	Latest	Containerized deployment
Markdown/JSON Export	Custom	Conversation history export

4 Key Learnings

- 100% local deployment is non-negotiable for semiconductor companies — validation logs contain proprietary process data that cannot leave the network.
- instructor-large embeddings outperform generic embeddings (e.g., all-MiniLM) on technical semiconductor content because they handle domain acronyms and jargon.
- Source citations build engineer trust — showing which document chunk the answer came from lets them verify and correct the AI.
- 7 document formats (.txt, .log, .report, .csv, .xlsx, .pdf, .docx) cover 99% of what validation engineers actually use.
- Ollama Llama2 on a 32GB Mac achieves <5s responses — good enough for interactive use, no GPU cluster needed.
- Conversation history export (Markdown/JSON) creates shareable investigation trails — critical for cross-team debugging sessions.